ENVIRONMENTAL ENGINEERING LAB

(BCE23502P)

LABORATORY MANUAL



DEPARTMENT OF CIVIL ENGINEERING

GIRIJANANDA CHOWDHURY UNIVERSITY

Hathkhowapara, Azara, Guwahati – 781017, Assam

SYLLABUS

Course Code	Course Title	L - T - P	Credit
BCE23502P	**Environmental Engineering Lab**	0 - 0 - 2	1

Pre-requisite: Nil

Course Objectives

The objective is to develop practical skills in analyzing key physical and chemical parameters of water and wastewater, and to introduce the measurement of ambient noise, fostering an understanding of their significance in environmental quality assessment and management.

Expected Course Outcome

Upon completion of this course, the student will be able to:

1. Accurately measure and interpret physical water quality parameters such as turbidity, electrical conductivity, pH, and various types of solids.
2. Perform standardized tests for alkalinity, acidity, hardness, Chemical Oxygen Demand (COD), Dissolved Oxygen (DO), and Biochemical Oxygen Demand (BOD), and understand their relevance to water pollution.
3. Determine the optimum coagulant dose for water treatment, demonstrating an understanding of coagulation-flocculation principles.
4. Utilize appropriate instrumentation to measure ambient noise levels and interpret the results in the context of environmental standards.

Test:1	Physical Characterization of water – Turbidity, Electrical Conductivity, pH	2 hours
Test:2	Analysis of solids content of water	3 hours
Test:3	Physical Characterization of water – Alkalinity and acidity, Hardness, Chloride	3 hours
Test:4	Optimum coagulant dose	2 hours
Test:5	Chemical Oxygen Demand (COD)	2 hours
Test:6	Biochemical Oxygen Demand (BOD)	3 hours
Test:7	Ambient noise measurement	1 hour
Total Lecture hours		**16 hours**

Mode of Assessment: Continuous Assessment Test, Assignments, Final Assessment Test.

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EXPERIMENT NO: 1

Determination of pH of a Water Sample

AIM: To determine the pH of a given water sample using a digital pH meter.

Instruments Required

- Digital pH meter
- Standard buffer solutions (pH 4.0, 7.0, 9.2 or 10.0)
- Distilled water
- Beakers (100 mL)
- Water samples (tap, river, pond, etc.)
- Wash bottle
- Tissue paper or clean cloth

Theory

The pH of water indicates its hydrogen ion concentration and is a measure of its acidity or alkalinity. It is a dimensionless number ranging from 0 to 14:

- pH < 7: Acidic
- pH = 7: Neutral
- pH > 7: Alkaline (Basic)

The pH of water affects chemical reactions, biological processes, and the solubility of nutrients and heavy metals. It is a critical parameter in water quality analysis for drinking, industrial, and environmental purposes.

$$pH = -\log[H^+]$$

In environmental engineering, maintaining appropriate pH levels is essential for wastewater treatment, corrosion control, and aquatic life health.

Procedure

A. Calibration of the pH Meter

1. Switch on the pH meter and allow it to warm up for a few minutes.
2. Rinse the electrode with distilled water and gently blot dry with tissue paper.
3. Immerse the electrode in the pH 7.0 buffer solution.
4. Adjust the meter to read 7.0 (follow the instrument manual).
5. Repeat the process using pH 4.0 and pH 9.2/10.0 buffers to complete calibration.

B. Measurement of Sample pH

1. Rinse the electrode with distilled water and blot dry.
2. Immerse the electrode into the sample water in a clean beaker.

3. Stir gently and allow the reading to stabilize.
4. Record the pH value.
5. Rinse the electrode before testing the next sample.

Precautions

1. Always calibrate the pH meter before use.
2. Do not touch the glass bulb of the electrode.
3. Rinse and blot the electrode between samples to avoid contamination.
4. Use fresh buffer solutions for accurate calibration.
5. Do not leave the electrode dry; store it in electrode storage solution when not in use.

Observation Table

Sl. No.	Sample Description	1	2	3	Avg. pH
1	Tap water				
2	Pond water				

Permissible Limits

As per BIS (IS 10500:2012) and WHO guidelines, the acceptable pH range for drinking water is 6.5 to 8.5.

Result

The average pH of the given water sample(s) is found to be:

- Tap Water: _____
- Pond Water: _____

EXPERIMENT NO: 2

Determination of Turbidity of a Water Sample

AIM: To determine the turbidity of a given water sample using a Nephelometric Turbidity Meter.

Apparatus

- Nephelometric Turbidity Meter
- Sample water
- Beakers
- Glass test tubes
- Distilled water (for rinsing)
- Clean lint-free cloth or tissue (for wiping cuvettes)

Theory

Turbidity is the measure of the relative clarity of a liquid. It is caused by the presence of suspended and colloidal matter such as clay, silt, finely divided organic and inorganic matter, plankton, and other microscopic organisms. Turbidity is an important parameter in water quality assessment. It affects light penetration and thus influences photosynthesis in aquatic ecosystems. Turbidity is measured in NTU (Nephelometric Turbidity Units) or JTU (Jackson Turbidity Units). The most commonly used method is nephelometry, where the intensity of light scattered at 90° by the suspended particles is measured.

Procedure

A. Calibration of Turbidity Meter

1. Turn on the turbidimeter and allow it to warm up (if required).
2. Clean the sample vial with distilled water and wipe it dry.
3. Fill the vial with standard turbidity solution (e.g., 0 NTU, 20 NTU, 100 NTU).
4. Insert the vial into the meter and adjust the calibration settings as per the device manual.
5. Repeat the process for other standard solutions to ensure accurate calibration.

B. Measurement of Sample Turbidity

1. Take the water sample in a clean vial.
2. Wipe the outer surface of the vial with a lint-free cloth to remove any fingerprints or droplets.
3. Insert the vial into the turbidimeter and close the lid.
4. Record the turbidity reading displayed in NTU.
5. Repeat the measurement 2–3 times for accuracy and record the average value.

Precautions

1. Always use clean and dry vials to avoid contamination.
2. Avoid touching the clear part of the vial with fingers.
3. Mix the sample well before taking measurements.
4. Calibrate the instrument before use with standard solutions.
5. Protect the sample from exposure to direct light during measurement.

Observation and Calculation

Sl. No.	Sample Description	1	2	3	Avg. Turbidity (NTU)
1	Tap water				
2	Pond water				

Result

The turbidity of the given water sample was found to be _____ NTU.

Permissible Limits: As per BIS (IS 10500:2012) and WHO guidelines, the desirable limit of turbidity in drinking water is:

- < 1 NTU (desirable)
- 5 NTU (maximum permissible, in absence of an alternative source)

EXPERIMENT NO: 3

Determination of Electrical Conductivity of a Water Sample

AIM: To determine the electrical conductivity (EC) of a given water sample using a digital conductivity meter.

Apparatus Required

- Digital conductivity meter
- Conductivity cell/probe
- Beakers (100 mL)
- Distilled water
- Standard potassium chloride (KCl) solution (for calibration)
- Water samples (tap, borewell, river, etc.)
- Tissue paper/cloth

Theory

Electrical Conductivity (EC) measures a water sample's ability to conduct electricity. This ability is directly related to the concentration of dissolved ions (salts) in the water, such as sodium, calcium, chloride, and sulfate. Therefore, EC is an indirect measure of the Total Dissolved Solids (TDS) in water.

- Unit: µS/cm (microsiemens per centimeter) or mS/cm (millisiemens per centimeter)

Relationship:

$$\text{TDS (mg/L)} \approx k \times \text{EC (µS/cm)}, \quad \text{where } k = 0.5\text{–}0.9$$

Factors affecting EC:

- Temperature (increases EC)
- Ion type and concentration

Procedure

A. Calibration of Conductivity Meter

1. Switch on the digital conductivity meter.
2. Rinse the probe with distilled water and blot dry.
3. Dip the probe into the standard KCl solution (usually 1413 µS/cm).
4. Adjust the meter reading to match the standard value.
5. Rinse the probe before testing the sample.

B. Measurement of Sample Conductivity

1. Take the water sample in a clean beaker.
2. Rinse the probe with distilled water, blot gently.

3. Immerse the probe in the sample and allow the reading to stabilize.
4. Note the conductivity value in µS/cm.
5. Repeat for different samples and take the average.

Precautions

1. Calibrate the instrument properly before use.
2. Always rinse the probe with distilled water between samples.
3. Avoid touching the tip of the probe.
4. Ensure the sample is at room temperature (or use automatic temperature compensation, if available).
5. Stir the sample gently before measurement.

Observation and Calculation

Sl. No.	Sample Description	1 (µS/cm)	2 (µS/cm)	3 (µS/cm)	Avg. EC (µS/cm)
1	Tap water				
2	Pond water				

Permissible Limits

As per BIS (IS 10500:2012) guidelines for drinking water:

- Desirable EC: Below 750 µS/cm
- Maximum permissible: Up to 2000 µS/cm (in absence of an alternate source)

Result

The average electrical conductivity of the water samples is:

- Tap Water: _____ µS/cm
- Pond Water: _____ µS/cm

EXPERIMENT NO: 4

Determination of Solids Content of a Water Sample

AIM: To determine the solids content of a given water sample, including Total Solids (TS), Total Dissolved Solids (TDS) and Total Suspended Solids (TSS).

Apparatus Required

- Drying oven (103–105°C)
- Crucibles or evaporating dishes (pre-weighed)
- Muffle furnace (for VS/FS determination – optional)
- Analytical balance (sensitivity 0.001 g)
- Filtration apparatus (filter paper or membrane filter)
- Desiccator
- Water bath (optional)
- Measuring cylinder or pipette
- Beakers
- Water sample
- Distilled water
- Filter paper

Theory

Solids in water are classified as:

- **Total Solids (TS):** All matter remaining as residue upon evaporation and drying at 103–105°C.
- **Total Suspended Solids (TSS):** Portion of solids retained by a filter (typically 1.5 μm).
- **Total Dissolved Solids (TDS):** Portion of solids passing through the filter and remaining after evaporation.
- **Volatile Solids (VS):** Organic fraction of solids lost upon ignition at 550°C.
- **Fixed Solids (FS):** Inorganic fraction remaining after ignition.

Excessive solids affect water quality, turbidity and aquatic life, and interfere with disinfection and industrial uses.

Procedure

A. Total Solids (TS)

1. Take a clean, dry crucible or dish. Weigh it accurately (W_1).
2. Add a known volume of the water sample (e.g., 100 mL).
3. Place the dish in a drying oven at 103–105°C for at least 1 hour.
4. Cool in a desiccator and weigh (W_2).

5. Calculate Total Solids using the formula below.

$$\text{Total Solids (mg/L)} = (W_2 - W_1) \times 1000 / \text{Volume of sample (mL)}$$

B. Total Dissolved Solids (TDS)

1. Filter the water sample using Whatman filter paper (or membrane filter).
2. Take a known volume of filtrate in a clean crucible. Weigh it (W_1).
3. Evaporate and dry at 180°C (or 103–105°C, as per standard method).
4. Cool in a desiccator and weigh (W_2).

$$\text{TDS (mg/L)} = (W_2 - W_1) \times 1000 / \text{Volume of filtrate (mL)}$$

C. Total Suspended Solids (TSS)

$$\text{TSS} = \text{TS} - \text{TDS}$$

Precautions

1. Always handle crucibles with tongs.
2. Ensure the balance is calibrated and accurate.
3. Do not over-dry or burn the sample.
4. Use clean, dry filter paper and glassware.
5. Cool samples in a desiccator to prevent moisture absorption before weighing.

Observation and Calculation

Sl. No.	Parameter	Value
1	Volume of sample (mL)	
2	Weight of empty dish (W_1) (g)	
3	Weight after drying (W_2) (g)	
4	Total Solids (mg/L)	
5	TDS (mg/L)	
6	TSS (mg/L)	

Permissible Limits

As per BIS (IS 10500:2012), for TDS (drinking water):

- Desirable: 500 mg/L
- Maximum permissible: 2000 mg/L

No specific limit is prescribed for TSS, but it should ideally be minimal for potable water.

Result

- Total Solids (TS): _____ mg/L
- Total Dissolved Solids (TDS): _____ mg/L
- Total Suspended Solids (TSS): _____ mg/L

EXPERIMENT NO: 5

Estimation of Optimum Coagulant Dose by Jar Test Apparatus

OBJECTIVE: To determine the optimum coagulant dosage for clarifying the given sample of water by using alum as the coagulant and performing the jar test experiment.

Theoretical Background

Introduction

Jar test is a simple device used to determine the optimum coagulant dose required. Jar testing is a method of simulating a full-scale water treatment process, providing system operators a reasonable idea of the way a treatment chemical will behave and operate with a particular type of raw water.

The jar test device consists of several stirrers (4–6) provided with pedals. The pedals can be rotated at varying speeds with the help of a regulator and motor. Samples are taken in the jars, and varying dosages of coagulant are added simultaneously to all the jars.

Colloids

Suspended solids with a diameter of less than one micron that cannot be removed directly by sedimentation are called colloids.

Colloids are so small in size that they cannot be settled under the action of gravity and carry a negative charge. Moreover, electrostatic forces of repulsion prevent them from flocculating. Therefore, a positively charged chemical (coagulant) is added to neutralize them, after which they get settled down.

Coagulation

The process of destabilization and aggregation of fine suspended solids by the addition of a chemical (polyelectrolyte) is called coagulation.

Coagulant

A chemical compound that is added to destabilize and bind colloids together is called a coagulant.

Flocculation

Slow mixing process that promotes agglomeration of colloids is called flocculation.

Advantages

Coagulants are used in water treatment plants:

- To remove suspended and colloidal particles.
- To assist the filtration process.
- To remove material that does not settle through direct sedimentation.

Disadvantages

- High cost of chemicals.
- Large amount of sludge is produced as a result of this process.
- This method is not economical; therefore, it is avoided for use at a large scale.

Significance

By performing jar tests, alternative treatment doses and strategies can be tried without altering the performance of the full-scale treatment plant, allowing easy comparison of the results of several different chemical treatments for time of floc formation, floc size, settleability and filtration characteristics.

Another important reason to perform jar testing is to save money. One of the common problems in water treatment is overfeeding or overdosing, especially with coagulants. This may not hurt the quality of water, but it can cost a lot of money. One of the easiest things an operator can do for optimization of the plant is jar testing, and jar testing is a must when looking at best available technologies.

Optimum Dose Measurement – Principle

Coagulants are used in water treatment plants:

- To remove natural suspended and colloidal matter.
- To remove material that does not settle in plain sedimentation, and to assist in filtration.

Alum [$Al_2(SO_4)_3.18H_2O$] is the most widely used coagulant. When alum solution is added to water, the molecules dissociate to yield SO_4^{2-} and Al^{3+}.

The positive species combine with negatively charged colloidal particles to neutralize part of the charge on the colloidal particle. Thus, agglomeration takes place.

Coagulation is a fairly complex phenomenon, and the coagulant should be distributed uniformly throughout the solution. This is achieved by:

- **Flash mixing:** The pedals are rotated at 100 or 120 rpm for 1 minute; this is called flash mixing.
- **Slow mixing:** The speed is reduced to 20 rpm for 20 minutes; this is called slow mixing. After a 30-minute settling time, supernatant is carefully drawn from all the jars to measure turbidity. The dosage that gives the least turbidity is taken as the optimum coagulant dose.

Sampling, Collection and Storage

- Take about 10 litres of free surface water, or prepare water by mixing soil into clean water to obtain a turbidity of up to 25 NTU.

Materials Required

Reagents

- **Aluminium Sulphate Solution** ($Al_2(SO_4)_3.18H_2O$): Take 2.5 g of aluminium sulphate in a 250 mL flask and dilute up to the mark with distilled water. (1 mL = 10 mg $Al_2(SO_4)_3$)
- **Sodium Carbonate Solution:** Dissolve 1.33 g of sodium carbonate in a 250 mL flask and dilute up to the mark with distilled water.
- **Sulphuric Acid Solution**
- **Sodium Hydroxide Solution**

Apparatus

- Jar test apparatus with 6 plastic jars.
- pH meter

- Turbidity meter
- Thermometer

Procedure

1. Measure the pH of the sample and bring it near neutral using sodium hydroxide or sulphuric acid solution.
2. Measure the turbidity of the sample as well.
3. Take equal volumes of sample, i.e., 1000 mL, into each of the 6 jars.
4. Check the operation of the stirrer; its pedals should be cleared of the jar walls by ¼ inch and 1 cm above the bottom.
5. Record the temperature of the sample in the jars at the start of the test.
6. Add the coagulant (aluminium sulphate and sodium carbonate solution) to 5 jars as given in the table below, leaving 1 jar without any coagulant, which serves as a control.

Jars	1	2	3	4	5	6
Vol. of $Al_2(SO_4)_3$ (mL)	2	4	6	8	10	0
Vol. of Na_2CO_3 (mL)	2	4	6	8	10	0

7. Stir at 120 rpm in each jar for approximately 1 minute.
8. Reduce the speed to 20 rpm and run at this stage for 20 minutes.
9. After the slow-mix period, withdraw the pedals and observe the settling of the particles. Allow the flocs to settle for 30 minutes; then withdraw a measured volume of supernatant liquor from mid-depth of the sample using a pipette, to determine turbidity with the turbidity meter.
10. Also record sample pH and temperature in each jar.
11. Plot the graph between alum dosage (x-axis) and turbidity (y-axis). The dosage of alum that represents the least turbidity is the optimum coagulant dose.
12. Repeat the jar test experiment if satisfactory results are not obtained; this is done by trying another range of coagulant doses, i.e., by narrowing the range of dosage near the optimum.

$$Al_2(SO_4)_3.18H_2O + 3Na_2CO_3 + 3H_2O \longrightarrow 2Al(OH)_3 + 3Na_2SO_4 + 3CO_3 + 18H_2O$$

Observations and Calculations

Coagulant Dose (mL)	Turbidity (NTU)
0	9.97
2	5.91
4	8.05
6	2.91

Coagulant Dose (mL)	Turbidity (NTU)
8	3.29
10	4.05

Graph

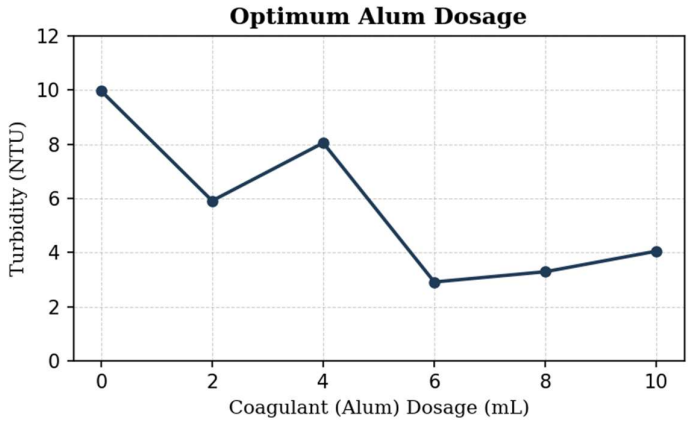


Fig. 5.1 – Turbidity vs. Coagulant (Alum) Dosage

Results

The graph shows that the optimum coagulant dosage of alum is 6 mg/L, corresponding to a turbidity of 2.91 NTU.

Comments

We performed this experiment in the lab and learnt many interesting and informative facts about the role of alum dosage, and coagulation and flocculation, in water treatment. It was a quite interesting and informative activity indeed.

EXPERIMENT NO: 6

Alkalinity Test

AIM: Determination of the alkalinity of a given water sample.

Theory

Alkalinity is determined by titrating the sample with a standard solution of a strong mineral acid to the bicarbonate and carbonic acid equivalence point. Alkalinity is expressed in terms of $CaCO_3$ equivalent. For samples whose pH is above 8.3, titration is done in two steps. In the first step, the pH is lowered to 8.3, which is indicated by the phenolphthalein indicator losing its pink colour and becoming colourless. In the second phase of titration, the pH is lowered to about 4.5, which is indicated by the methyl orange indicator changing colour from yellow to orange-red.

Apparatus Required

- Burette
- Pipette
- Erlenmeyer flask

Reagents

- Sulphuric acid – 0.02 N (H_2SO_4)
- Phenolphthalein indicator
- Methyl orange indicator

Procedure

1. A sample volume of 100 mL is taken into a clean flask, V.
2. One or two drops of phenolphthalein indicator are added. If the pH is above 8.3, the colour turns pink.
3. The sample is titrated against standard 0.02 N H_2SO_4 until the colour just disappears. The volume of titrant used is noted as V_1.
4. Then two drops of methyl orange indicator are added. The colour turns yellow.
5. The sample is again titrated against the acid in the burette until the yellow colour just turns orange-yellow. The volume of titrant used is noted as V_2.

Observation

Sl. No.	Volume of Sample (V)	Initial Burette Reading (mL)	Final Burette Reading (mL)	Volume of H_2SO_4 (mL)	Alkalinity (mg/L)
1					
2					
3					

Calculation

Phenolphthalein alkalinity, expressed as mg/L (as $CaCO_3$):

$$P = (V_1 \times 50 \times 1000 \times 0.02\ N) / \text{Volume of sample used (mL)}$$

Total alkalinity, expressed as mg/L (as $CaCO_3$):

$$T = (V_2 \times 50 \times 1000 \times 0.02\ N) / \text{Volume of sample used (mL)}$$

Result

- Phenolphthalein alkalinity, expressed as mg/L (as $CaCO_3$) = _____
- Methyl orange (total) alkalinity, expressed as mg/L (as $CaCO_3$) = _____

EXPERIMENT NO: 7

Acidity Test

AIM: To determine the mineral and phenolphthalein acidity of a given water sample.

Theory

Acids contribute to corrosiveness and influence chemical reaction rates, chemical speciation and biological processes. Acidity of water is its quantitative capacity to react with a strong base to a designated pH. The measured value may vary significantly with the end-point pH used in the determination.

When the chemical composition of a sample is known, mineral acids, weak acids such as carbonic and acetic acid, and hydrolyzing salts such as iron or aluminium sulphate may contribute to the measured acidity, according to the method of determination.

- **Mineral acidity:** Measured by titration to a pH of about 3.5, the methyl orange end point (also known as methyl orange acidity).
- **Total acidity:** Titration of a sample to the phenolphthalein end point of pH 8.3 measures mineral acidity plus acidity due to weak acids; thus it is called total acidity (or phenolphthalein acidity).

In water analysis, this test does not bear significant importance because methyl orange acidity invariably remains absent in raw water, and even phenolphthalein acidity (principally due to the excessive prevalence of dissolved carbon dioxide and carbonic acid) normally does not exist to a significant extent in raw water.

Apparatus Required

- pH meter

Reagents

- Sodium hydroxide titrant – 0.02 N (NaOH)
- Phenolphthalein indicator
- Methyl orange indicator

Procedure

1. Take 50 mL of sample in a conical flask and add 2–3 drops of methyl orange indicator solution.
2. Fill the burette with 0.02 N NaOH solution and titrate until the colour of the solution just changes to a faint orange colour, indicating the end point. Record the volume of titrant consumed as V_1 (mL). Calculate the methyl orange acidity using Eq. (1a):

$$\text{Methyl orange acidity (or Mineral Acidity)} = (V_1 \times 1000) / \text{Sample volume} \quad \textit{...(1a)}$$

When the 0.02 N NaOH solution used in titration is not standardized, mineral acidity is calculated using the following equation (1b):

$$\text{Methyl orange acidity} = (V_1 \times N \times 50 \times 1000) / \text{Sample volume} \quad \textit{...(1b)}$$

For the phenolphthalein acidity test, add 2–3 drops of phenolphthalein indicator to the water sample from step 2 and continue the titration until a faint pink colour develops in the solution (the end point of titration).

Record the volume of titrant consumed as V_2 (mL) and calculate total acidity, or phenolphthalein acidity, using Eq. (2):

$$\text{Total acidity (or Phenolphthalein Acidity)} = (V_2 \times N \times 50 \times 1000) / \text{Sample volume} \quad ...(2)$$

Observation

Sl. No.	Sample Volume (mL)	Titrant Used, V1 (mL)	Titrant Used, V2 (mL)	Acidity (mg/L)
1				
2				
3				

Result

- Methyl orange (mineral) acidity = _____ mg/L
- Phenolphthalein (total) acidity = _____ mg/L

Conclusion

The mineral acidity and phenolphthalein acidity of the given water sample are determined.

EXPERIMENT NO: 8

Hardness Test

AIM: To determine the hardness of a given sample of water by the EDTA titrimetric method.

Theory

Hard waters are generally considered to be those waters that require considerable amounts of soap to produce foam, and that also produce scale in water pipes, heaters, boilers and other units in which the temperature of water is increased. Hard water is appropriate for human consumption, similar to soft water; however, it produces adverse reactions with soap, and thus its use for cleaning purposes is unsatisfactory, requiring its removal from water. Hardness of water varies from place to place. In general, surface waters are softer than ground waters. Waters are commonly classified based on the degree of hardness (Table 1):

Hardness (mg/L)	Degree of Hardness
0–75	Soft
75–150	Moderately hard
150–300	Hard
> 300	Very hard

Table 1 – Classification of Hardness

When hardness is numerically greater than the sum of carbonate and bicarbonate alkalinity, that amount of hardness equivalent to the total alkalinity is called "carbonate hardness"; the amount of hardness in excess of this is called "non-carbonate hardness." When the hardness is numerically equal to or less than the sum of carbonate and bicarbonate alkalinity, all hardness is carbonate hardness, and non-carbonate hardness is absent. Hardness may range from zero to hundreds of milligrams per litre.

Apparatus Required

- Burette
- Pipette
- Conical flask

Reagents Required

- Buffer solution (pH 10)
- Standard EDTA titrant – 0.01 M
- Eriochrome Black T

Procedure

1. Take 50 mL of the sample in a conical flask.
2. Add 1 mL of buffer solution, then add 2 crystals of Eriochrome Black T.
3. Titrate with standard EDTA titrant until the colour changes to a distinct blue.

Calculation

Hardness (EDTA), as mg $CaCO_3$/L = (A × B × 1000) / mL of sample = (Titrant volume (mL) × 1000) / Volume of sample (mL)

Where,

- A = mL of titration of sample
- B = mg $CaCO_3$ equivalent to 1.00 mL EDTA titrant

Observation Table

Sl. No.	Quantity of Sample Taken (mL)	Initial Burette Reading (A)	Final Burette Reading (B)	mL of EDTA Used (B−A)	Hardness (mg $CaCO_3$/L)
1					
2					
3					

Result

Hardness of the given water sample = _____ mg $CaCO_3$/L

Conclusion

The hardness of the given water sample is determined by the EDTA titrimetric method.

EXPERIMENT NO: 9

Chloride Test

AIM: To determine the concentration of chloride in a given water sample.

Theory

Chloride occurs in all natural water in widely varying concentrations. It is not harmful to humans if the concentration is less than 250 mg/litre.

Chloride can be readily measured by the use of 0.014 N silver nitrate, by the Argentometric method. This method is equivalent to 0.5 mg of chloride ion for the titration of 1 mL of this silver nitrate solution.

The end point can be detected by K_2CrO_4 indicator, in the presence of excess Ag^+. The indicator normally used is potassium chromate, which supplies the chromate ion. As the concentration of chloride ion approaches extinction, the silver ion concentration increases to a level at which the solubility product of silver chromate is exceeded, and it begins to form a reddish-brown precipitate.

Reaction:

$$2Ag^+ + CrO_4^{2-} \longrightarrow Ag_2CrO_4$$

Apparatus Required

- Burette
- Pipette
- Measuring cylinder
- Beaker

Reagents

- Potassium chromate indicator solution (K_2CrO_4)
- Standard silver nitrate titrant (0.014 N $AgNO_3$)
- H_2SO_4 or NaOH

Procedure

1. Take 100 mL of sample in a beaker.
2. Adjust the sample pH to 7–10 with H_2SO_4 or NaOH, if it is not already within this range.
3. Add 1 mL of potassium chromate indicator solution.
4. Titrate with standard $AgNO_3$ titrant to a pinkish-yellow end point.
5. Titrate a reagent blank with standard $AgNO_3$ titrant by the same method, and establish the blank value.

Observation

Sl. No.	Quantity of Sample (mL)	Initial Burette Reading (A)	Final Burette Reading (B)	mL of $AgNO_3$ Used (B−A)	Chloride (mg/L)
1					
2					
3					

Calculation

$$\text{Chloride, as mg } Cl^-/L = (A - B) \times N \times 35450 / \text{mL of sample}$$

Where,

- A = mL of titration for sample
- B = mL of titration for blank
- N = Normality of $AgNO_3$

Result

Chloride concentration of the given water sample = _____ mg/L

Conclusion

The concentration of chloride in the given water sample is determined by the Argentometric method.

EXPERIMENT NO: 10

COD Test

AIM: Determination of the COD value of a given water sample.

Theory

Chemical Oxygen Demand (COD) is the amount of a specific oxidizing agent that reacts with a sample under controlled conditions, expressed as oxygen equivalence. This parameter indicates the extent of organic matter contamination of water and is always higher than the Biochemical Oxygen Demand (BOD). It is used to indicate organic matter contamination and helps in assessing the overall organic load reaching the receiving body.

Selection of Method

There are two methods for COD determination. The *open reflux method* is suitable for a wide range of wastes where a large volume of sample is required (for samples with COD ≥ 50 mg O_2/L). In the *closed reflux method*, small quantities of metallic salt reagents are required and small quantities of hazardous waste are produced (for samples with COD = 5 to 50 mg O_2/L). In the closed reflux method, ampules and culture tubes with pre-measured reagents are used, and the sample is placed in the tube for COD determination.

In this experiment, the closed reflux method is used, and samples with COD < 50 mg O_2/L are tested.

Reaction with Dichromate Solution of Sample

Potassium dichromate is a strong oxidizing agent and can be used to prepare a solution of exact normality.

$$C_nH_aO_bN_c + d\ Cr_2O_7^{2-} + (8d+c)\ H^+ \longrightarrow n\ CO_2 + [(a+8d-3c)/2]\ H_2O + c\ NH_4^+ + 2d\ Cr^{3+} \quad ...(1)$$

Here, $d = (2n/3) + (a/6) - (b/3) - (c/2)$

During the experiment, the excess dichromate concentration is determined by titrating it with ferrous ammonium sulphate (FAS). The reaction is given by:

$$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O \quad ...(2)$$

Ferroin (ferrous 1,10-phenanthroline sulphate) is used to indicate the change in oxidation–reduction potential of the solution; it indicates the condition when all dichromate has been reduced by ferrous ion. It gives a very sharp brown colour change, which can be seen despite the blue-green colour generated by the Cr^{3+} ions formed on reduction of the dichromate.

Apparatus Required

1. Digestion vessels
2. Block heater
3. Microburette; ampule sealer
4. Borosilicate culture tubes (16 mm × 100 mm or 20 mm × 150 mm) with TFE-lined screw caps

The block heater is required to operate at 150 ± 2°C, with holes to accommodate digestion vessels. An oven must not be used, owing to the possibility of leaking samples generating a corrosive and explosive atmosphere.

Reagents

- **Standard potassium dichromate digestion solution, 0.01667 M:** Add to about 500 mL distilled water 4.903 g $K_2Cr_2O_7$ (primary standard grade, previously dried at 150°C for 2 h), 167 mL conc. H_2SO_4, and 33.3 g $HgSO_4$. Dissolve, cool to room temperature, and dilute to 1000 mL.
- **Sulphuric acid reagent.**
- **Ferroin indicator solution:** Dilute by a factor of 5 as required; used to indicate the change in oxidation–reduction potential of the solution.
- **Standard ferrous ammonium sulphate titrant (FAS), ≈ 0.10 M:** Dissolve 39.2 g $Fe(NH_4)_2(SO_4)_2.6H_2O$ in distilled water. Add 20 mL conc. H_2SO_4, cool, and dilute to 1000 mL. Standardize the solution daily against the standard $K_2Cr_2O_7$ digestion solution as follows: pipette 5.00 mL digestion solution into a small beaker, add 10 mL reagent water to substitute for the sample, cool to room temperature, add 1–2 drops of diluted ferroin indicator and titrate with FAS titrant.

$$\text{Molarity of FAS solution} = [V_{K2Cr2O7} \times 0.1] / V_{FAS} \quad ...(3)$$

Where, $V_{K2Cr2O7}$ = volume of $K_2Cr_2O_7$ (mL); V_{FAS} = volume of FAS (mL)

- **Sulphamic acid.**
- **Potassium hydrogen phthalate standard.**

Procedure

1. Wash culture tubes and caps with 20% H_2SO_4 before use, to prevent contamination.
2. Place the sample in a culture tube or ampule and add digestion solution. Carefully run sulphuric acid reagent down the inside of the vessel so an acid layer forms under the sample-digestion solution layer; tightly cap tubes or seal ampules, and invert each several times to mix completely.
3. Place tubes or ampules in a block digester preheated to 150°C and reflux for 2 h, behind a protective shield. CAUTION: these sealed vessels may be under pressure from gases generated during digestion; wear face and hand protection, as dangerous pressures will be generated at 150°C.
4. Cool to room temperature and place the vessels in a test tube rack. Some mercuric sulphate may precipitate out, but this will not affect the analysis.
5. Remove the culture tube caps and add a small TFE-covered magnetic stirring bar. If ampules are used, transfer contents to a larger container for titrating.
6. Add 0.05–0.10 mL (1–2 drops) of ferroin indicator and stir rapidly on a magnetic stirrer while titrating with standardized 0.10 M FAS. The end point is a sharp colour change from blue-green to reddish brown, although the blue-green may reappear within minutes. In the same manner, reflux and titrate a blank containing the reagents and a volume of distilled water equal to that of the sample.
7. COD is given by the formula below.

$$\text{COD, as mg } O_2/L = [(A - B) \times M \times 8000] / V_{sample} \quad ...(4)$$

Where, A = volume of FAS used for blank (mL); B = volume of FAS used for sample (mL); M = molarity of FAS; 8000 = milliequivalent weight of oxygen × 1000 mL/L.

EXPERIMENT NO: 11

BOD Test

AIM: Determination of the BOD of a given sample.

Theory

The most widely used test indicating organic pollution of both wastewater and surface water is the 5-day BOD (BOD_5). This determination involves measuring the dissolved oxygen used by microorganisms in the biochemical oxidation of organic matter. BOD_5 is the total amount of oxygen consumed by microorganisms during the first five days of biodegradation. Oxygen demand is associated with the biodegradation of the carbonaceous portion of wastes and the oxidation of nitrogen compounds such as ammonia. The following equations simplify the biodegradation process:

$$\text{Organic matter} + O_2 + \text{microorganisms} \longrightarrow CO_2 + H_2O + \text{new microbial cells}$$

$$\text{Ammonia} + O_2 + \text{microorganisms} \longrightarrow NO_3^- + H_2O + \text{new microbial cells}$$

Apparatus Required

- Incubation bottle (300 mL volume)
- Air compressor
- 20°C incubator

Reagents Required

- **Manganese sulphate solution:** Dissolve 480 g $MnSO_4.4H_2O$, 400 g $MnSO_4.2H_2O$, or 364 g $MnSO_4.H_2O$ in distilled water, filter, and dilute to 1 L. The $MnSO_4$ solution should not give a colour with starch when added to an acidified potassium iodide (KI) solution.
- Alkali-iodide-azide reagent.
- Sulphuric acid: one mL is equivalent to ~3 mL alkali-iodide-azide reagent.
- Starch solution: dissolve 2 g laboratory-grade soluble starch and 0.2 g salicylic acid, as preservative, in 100 mL hot distilled water.
- **Standard sodium thiosulphate titrant:** dissolve 6.205 g $Na_2S_2O_3.5H_2O$ in distilled water, add 1.5 mL 6N NaOH or 0.4 g solid NaOH, and dilute to 1000 mL. Standardize with biiodate solution.
- **Standard potassium biiodate solution (0.0021 M):** dissolve 812.4 mg $KH(IO_3)_2$ in distilled water and dilute to 1000 mL.
- **Standardization:** dissolve ~2 g KI, free from iodate, in an Erlenmeyer flask with 100–150 mL distilled water; add 1 mL 6N H_2SO_4 (or a few drops of conc. H_2SO_4) and 20.00 mL standard biiodate solution. Dilute to 200 mL and titrate the liberated iodine with thiosulphate titrant, adding starch toward the end of titration, when a pale straw colour is reached. When the solution is exactly equal, 20.00 mL of 0.025 M $Na_2S_2O_3$ should be required; if not, adjust the $Na_2S_2O_3$ solution to 0.025 M.

Procedure

DO Measurement

1. Prepare dilution water by adding 2 mL/L of the following reagents to distilled water: (a) phosphate buffer solution, (b) magnesium sulphate solution, (c) calcium chloride solution, (d) ferric chloride solution, (e) sodium sulphite solution.
2. For a given sample bottle, add 1 mL of alkali-iodide-azide reagent and then 1 mL of manganous sulphate solution. Shake the bottle well and keep it open for 5 minutes to allow the precipitate to settle. Add 2 mL concentrated H2SO4 and cap the bottle. Shake well until all the precipitate has dissolved.
3. Take 200 mL of sample in a conical flask and titrate with standard sodium thiosulphate solution (0.025N) until the colour changes from dark yellow to light yellow. Then add a few drops of starch indicator and continue titrating until the solution becomes colourless or returns to its original sample colour. Note the volume of 0.025N sodium thiosulphate consumed.
4. Calculate the DO value of the sample. In a 200 mL sample, 1 mL of 0.025N sodium thiosulphate is equivalent to 1 mg/L dissolved oxygen:

   Dissolved Oxygen (DO) (mg/L) = mL of sodium thiosulphate (0.025N) consumed

BOD

1. Prepare BOD dilutions using dilution water (containing nutrients), e.g.: 10 mL sample in a 300 mL BOD bottle, filled up with dilution water; 15 mL sample in a 300 mL BOD bottle, filled up with dilution water; 20 mL sample in a 300 mL BOD bottle, filled up with dilution water.
2. Take 300 mL sample in a BOD bottle. Prepare two sets of this sample: keep one set for DO analysis on day 0 (Sample 0-Day), and incubate the other in the BOD incubator for 5 days at 20°C (Sample 5-Day).
3. Measure DO in the different samples at t = 0.
4. Incubate the samples at 20°C for 5 days.
5. Return to the lab after 5 days and record the dissolved oxygen.
6. Record the data as shown in the table below.

Bottle No.	Wastewater Sample (mL)	Initial DO, DO_0 (mg/L)	DO at 5 Days, DO_5 (mg/L)

The 5-day BOD value of the sample at 20°C is calculated as:

$$\text{t-day BOD} = [DO_t - DO_0] / P \quad \textit{...(1)}$$

where P = Dilution factor = 300 mL / (sample volume in mL)

EXPERIMENT NO: 12

Measurement of Ambient Noise Level

AIM: To measure the ambient noise level (sound pressure level) at a selected location, compute the equivalent continuous sound level (L_{eq}), compare it with applicable standards (e.g., Central Pollution Control Board (CPCB) standards), and analyse the spatial/temporal variation of noise in different zones (industrial, commercial, residential, silence) in accordance with the Environmental Engineering Laboratory syllabus.

Theory

- Sound is a pressure wave in air, which, when unwanted, is called noise.
- The common metric is the sound pressure level (SPL), in decibels (dB).
- Human ear sensitivity varies with frequency; thus, sound level meters use weighting networks such as A-, B-, C-weighting (the most common being A-weighting).
- Sound measurement is influenced by microphone placement, obstacles, reflecting surfaces, background noise, etc.
- Standards: In India, the CPCB prescribes ambient noise level limits for different zones and day/night times.

Apparatus

- Sound level meter (Type 1 or Type 2), with A-weighting and time-weighting options (Fast/Slow).
- Tripod (optional), for stable mounting of the microphone.
- Calibrator (for pre-/post-calibration of the sound level meter).
- Notebook or data-logging facility (some sound meters can record data).
- Measuring tape/markers, to establish location distances.
- Stopwatch.

Location and Sampling Plan

- Select locations in different zones (industrial, commercial, residential, silence zone).
- Choose measurement points away from obstacles, reflective surfaces and facades, where possible. Place the microphone at a height of ~1.2–1.5 m above ground.
- Decide measurement durations, e.g., measure for 1 minute (or more) at fixed intervals at each location, replicating 2–3 times.
- Record the time of day (day/night), weather conditions and any interfering noise events.
- Ensure background noise and any dominating noise source are noted (e.g., traffic, machinery), so they can be accounted for or excluded, if necessary.

Procedure

1. Calibration: Before measurement begins, calibrate the sound level meter using the calibrator at the reference level (e.g., 94 dB at 1 kHz) and note down the calibration condition.

2. At the first location, carry out the sub-steps below.
3. After completing measurements at all locations, perform a post-calibration check (to ensure the instrument has not drifted).
4. Compute the required results, e.g., average SPL and L_{eq} for each location, and compare across zones using the specific formula.
5. Compare the results with CPCB ambient noise standards (see table below).

At the first location (Step 2):

- Set the weighting to A-weighting (dB(A)).
- Set the time weighting (usually Slow, for ambient noise).
- Select the appropriate measurement range on the instrument (e.g., 30–90 dB).
- Mount or hold the microphone at ~1.2 m height, oriented toward the general noise field, not directly at a source, unless specified.
- Measure for, e.g., 1 minute continuously; note the instantaneous SPL readings, or let the meter compute L_{eq} if it has that feature.
- Record the location, time, weather conditions, and measurement readings (max, min, average, if available).
- Repeat for the next locations.

CPCB Ambient Noise Standards

Zone	Day Limit, Leq (dB(A))	Night Limit, Leq (dB(A))
Industrial	75	70
Commercial	65	55
Residential	55	45
Silence Zone	50	40

